Hallucinations and Key Information Extraction in Medical Texts: A Comprehensive Assessment of Open-Source Large Language Models

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***Abstract*—Clinical summarization is crucial in healthcare as it distills complex medical data into digestible information, enhancing patient understanding and care management. Large language models (LLMs) have shown significant potential in automating and improving the accuracy of such summarizations due to their advanced natural language understanding capabilities. These models are particularly applicable in the context of summarizing medical/clinical texts, where precise and concise information transfer is essential. In this paper, we investigate the effectiveness of open-source LLMs in extracting key events from discharge reports, such as reasons for hospital admission, significant in-hospital events, and critical follow-up actions. In addition, we also assess the prevalence of various types of hallucinations in the summaries produced by these models. Detecting hallucinations is vital as it directly influences the reliability of the information, potentially affecting patient care and treatment outcomes. We conduct comprehensive numerical simulations to rigorously evaluate the performance of these models, further probing the accuracy and fidelity of the extracted content in clinical summarization.**

***Index Terms*—Medical Text Summarization, Large Language Models, Hallucinations, Key Information Extraction, Factual Accuracy.**

I. Introduction

Clinical text summarization [1] is a crucial task in modern healthcare, as it enables efficient extraction of key medical information from lengthy and complex documents such as electronic health records (EHRs), discharge summaries, and radiology reports. The vast amount of unstructured textual data generated in clinical settings poses a significant challenge for healthcare professionals, who must rapidly interpret patient histories, diagnoses, and treatment plans to make informed decisions [2]. Effective summarization of medical texts can enhance clinical workflow efficiency and improve patient outcomes by ensuring that essential information is readily accessible [3]. Additionally, it plays a vital role in medical research, enabling quicker literature reviews and knowledge synthesis [4]. However, traditional summarization techniques often struggle with the specialized language, domain-specific jargon, and contextual nuances of medical texts, highlighting the need for more advanced, AI-driven approaches [5].

Large Language Models (LLMs) have emerged as powerful tools in artificial intelligence and machine learning, demonstrating remarkable capabilities in natural language understanding, generation, and contextual reasoning [6]. These models, trained on vast and diverse datasets, can comprehend and generate human-like text, making them particularly well-suited for tasks such as summarization, translation, and question-answering [7]. In the medical domain, LLMs offer significant potential for summarizing clinical texts [8] by capturing key information while preserving critical details necessary for decision-making. Unlike traditional rule-based or statistical summarization methods, LLMs can adapt to complex linguistic structures, recognize medical terminologies, and generate concise yet informative summaries tailored to specific clinical needs. By leveraging contextual embeddings and domain-specific fine-tuning, LLMs can improve the accuracy and relevance of medical text summarization, ultimately supporting healthcare professionals in reducing information overload and enhancing patient care [6].

Despite their potential, LLMs face notable challenges in summarizing medical texts, which can impact the reliability and accuracy of generated summaries. These challenges include:

Key Information Extraction: Extracting clinically relevant events from medical texts is a critical requirement for summarization. However, LLMs may struggle to prioritize essential details, such as symptoms, treatments, and medication changes, especially when faced with lengthy or complex narratives. Without proper mechanisms for identifying and preserving crucial information, summaries may omit significant medical events, leading to incomplete or misleading interpretations.

Hallucinations: LLMs are known to generate plausible yet incorrect or non-existent information, a phenomenon referred to as hallucination [9], [10]. In the medical domain, hallucinated details in summary, such as incorrect dosages, fabricated conditions, or misrepresented patient histories, thus can have serious consequences for clinical decision-making. This challenge underscores the need for robust validation mechanisms, domain-specific fine-tuning, and external fact-checking to ensure the reliability of medical text summaries [11].

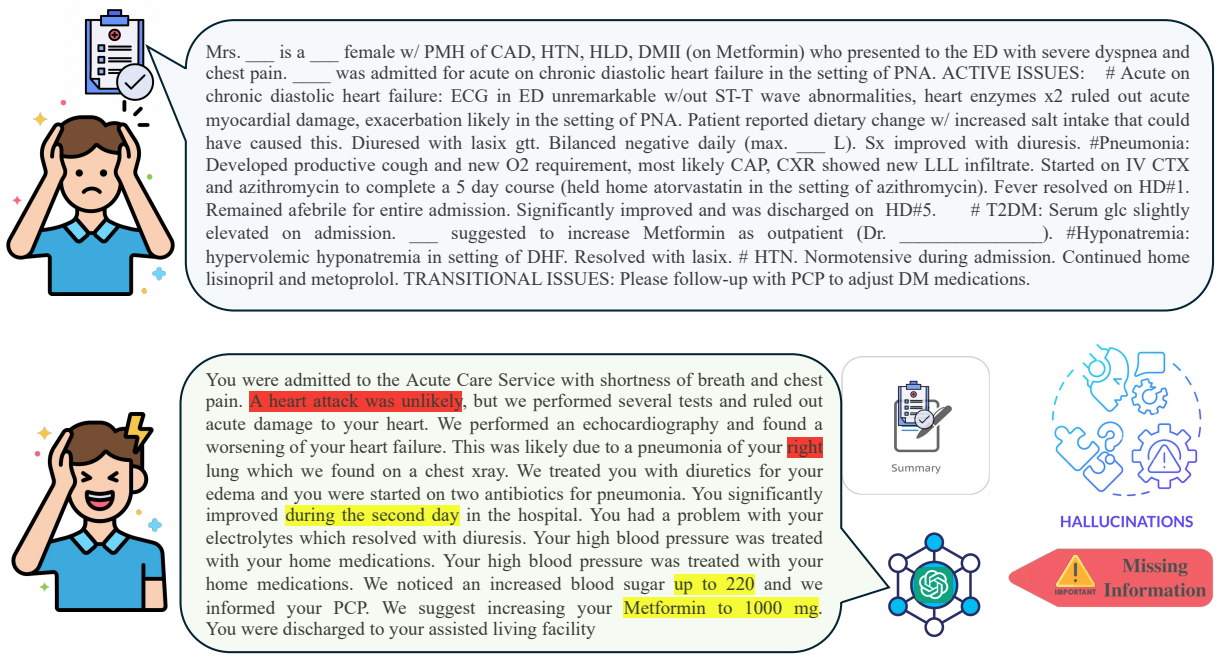

Fig. 1: A sample hospital discharge report (top) where the patient initially could not understand the content. However, by using an LLM, the patient is able to understand the summary (bottom) of the discharge report. Note that the LLM-generated summary may contain hallucinations, and key information may be missing. For instance, the yellow and red highlighted portions are unsupported and incorrect/contradicted fact hallucinations, respectively.

In this paper, we aim to highlight and analyze the challenges of key event extraction and hallucinations in LLM-driven medical text summarization. To achieve this, we conduct a comprehensive evaluation using discharge summaries from the MIMIC-IV dataset [12], a widely used repository of de-identified clinical records. We explore various open-source large language models (LLMs) to assess their effectiveness in extracting essential medical events while minimizing hallucinations. The open-source LLMs used in this study include LLaMA [13], Mistral [14], Gemma [15], Phi [16], Falcon [17], LLaVA [18], DeepSeek [19], and Qwen [20]. Our analysis focuses on understanding the strengths and limitations of these models in handling domain-specific language and preserving critical clinical information. Addressing these issues is essential to making LLM-driven summarization a practical and trustworthy tool for healthcare applications.

II. Prior Works and Our Contributions

Medical text summarization has been an active area of research, with numerous studies exploring techniques to extract essential clinical information while maintaining accuracy and reliability. Traditional NLP approaches have relied on rule-based and machine-learning methods, while recent advancements in deep learning and large language models have significantly improved the capabilities of automatic summarization [21]. However, despite these advancements, two key challenges remain particularly difficult: (1) accurately extracting key medical events and (2) addressing hallucinations in LLM-generated summaries. Addressing these issues is crucial for ensuring that AI-driven summarization tools can be safely and effectively deployed in healthcare settings.

A. Previous Works

Understanding the challenges of medical text summarization requires examining prior research on key event extraction and hallucination detection in LLM-generated summaries. In the following sections, we discuss existing works addressing these issues and their relevance to our study.

1) Key Event/Information Extraction: Extracting essential clinical information from lengthy medical documents has been a long-standing challenge in natural language processing. Traditional approaches relied on rule-based methods and statistical models, such as conditional random fields and hidden Markov models, to identify key events. More recently, deep learning-based techniques, including recurrent neural networks, convolutional neural networks, and transformer-based architectures [22], have been employed to improve accuracy. Pre-trained biomedical models such as BioBERT [23] and ClinicalBERT [24] have demonstrated effectiveness in extracting medical entities from structured and unstructured clinical narratives. However, these methods often require extensive domain-specific training data and struggle with generalization across different medical contexts.

With the advent of LLMs, researchers have explored their potential for key event extraction in clinical text summarization. Models such as GPT-4 [25] and Med-PaLM [26] have shown promise in capturing contextual dependencies, but they may still fail to prioritize clinically relevant insights accurately. Furthermore, without explicit fine-tuning on labeled clinical datasets, LLMs risk omitting critical patient information, making their direct application to medical summarization challenging.

2) Detection of Hallucination: A major limitation of LLMs in medical summarization is their tendency to generate hallucinated content, i.e., fabricated or misleading information that does not align with the input text. Hallucinations in medical AI can have serious consequences, potentially leading to incorrect diagnoses, erroneous treatment recommendations, and misinterpretation of patient histories.

Existing research has explored various techniques for hallucination detection and mitigation in NLP applications. Some approaches rely on fact-checking methods that compare generated summaries against source texts using similarity metrics or retrieval-based validation. Others employ uncertainty quantification techniques, such as confidence scoring and probabilistic modeling, to assess the reliability of LLM-generated outputs. In the biomedical domain, external knowledge bases (e.g., SNOMED CT [27]) have been integrated into NLP pipelines to verify the factual accuracy of generated content. However, ensuring faithfulness in LLM-generated medical summaries remains an open challenge, necessitating further research into robust evaluation metrics and hallucination filtering mechanisms.

B. Summary of Contributions

In this paper, we build upon existing research by conducting a comprehensive analysis of LLM-driven summarization of medical texts, focusing on key event extraction and hallucination detection. Our main contributions include:

- Evaluation of Open-Source LLMs for Clinical Summarization: We explore various publicly available large language models to assess their performance in summarizing discharge summaries from the MIMIC-IV dataset.
- Analysis of Key Event Extraction Capabilities: We investigate how well different LLMs identify and retain critical clinical insights, such as diagnoses, treatments, and medication changes.
- Hallucination Detection and Quantification: We evaluate the extent of hallucinations in LLM-generated summaries and analyze potential factors contributing to misleading or fabricated information.

By systematically examining these challenges, we aim to provide valuable insights into the feasibility of using LLMs for medical text summarization and propose directions for improving their reliability in clinical applications.

III. Analyzing LLM-Generated Clinical Summaries

In this section, we examine the key aspects of essential information extraction and corresponding hallucinations in clinical summarization by LLMs. To illustrate these, we provide a detailed analysis using a representative hospital discharge report as an example.

A. Key Event Extraction in Clinical Summarization

Text summarization is a key task in natural language processing, condensing extensive texts into concise summaries while preserving essential information. It is widely used across domains requiring swift data comprehension, such as finance, where it streamlines market reports and investment analyses by highlighting critical trends and risks, and law, where it simplifies legal documents to emphasize case facts, arguments, and verdicts. In medical applications, summarization extracts vital clinical details from documents like patient histories and treatment plans, aiding both healthcare professionals and patients. For example, summarizing a CT scan report can highlight key findings, facilitating swift clinical decisions. Leveraging open-source LLMs for medical text summarization enhances patient care by improving communication and clarity of critical information.

This study utilizes open-source LLMs to summarize hospital discharge reports, simplifying complex medical jargon for better patient comprehension. We assess their efficacy in generating patient-friendly summaries by extracting key details, including admission reasons, significant hospital events, and essential follow-up actions, analyzed through a specific example. Here, we consider an example of the hospital's discharge report presented in Fig. 2 to systematically extract and assess key clinical data. The methodology identifies admission reasons, key interventions, diagnostic events, and follow-ups, ensuring a clear understanding of the patient's healthcare journey and critical decisions.

In this example, the patient's admission is due to acute symptoms like fever, nausea, and vomiting, along with chronic conditions such as hypertension requiring urgent care. Hospitalization involves administering broad-spectrum antibiotics and conducting extensive diagnostic imaging. The discharge plan includes essential follow-ups, such as liver function tests and a chest CT in six months for ongoing monitoring. We tasked each LLM with generating a summary of this hospital discharge report, limiting their responses to a maximum of 1000 characters. Our goal is to evaluate whether the open-source LLMs effectively capture these critical details in their summaries. Fig. 3 presents some examples of such summaries produced by the LLMs.

Consider the summaries in Fig. 3(a) and 3(b) generated by Gemma2 and LLaMA. While both captured several key events, they overlooked critical details. Notably, neither summary included liver function monitoring or hypertension management in the follow-up plan. Additionally, Gemma2 failed to recognize the patient's existing esophageal cancer diagnosis. Despite being well below the 1000-character limit (505 and 617 characters, respectively), the summaries had enough space to include these essential details but failed to do so.

B. Hallucination in LLM-Generated Summaries

A major limitation of LLMs in medical applications is their tendency to generate hallucinated content—information that is not present in the source text or is factually incorrect [28]. In the context of clinical summarization, hallucinations can be particularly problematic, as they may introduce inaccurate medical details, misrepresent patient histories, or suggest incorrect treatments, posing risks to patient safety.

Thus, hallucinations in text generated by LLMs can manifest in various forms, including the following:

___ year old male with stage IIIB esophageal adenocarcinoma s/p neoadjuvant chemo/XRT prior to esophagectomy in ___ who presented with fevers, nausea, and vomiting. # Fever: He was treated broadly with vancomycin and cefepime in the ED. He had a CXR, UA and CT abdomen/pelvis that did not show a source of infection. He was felt to most likely have a viral gastroenteritis. … … … His shoulder lesion which had been positive on recent ___ was not painful and he had full ROM in his shoulder joint and it was felt to be unlikely to be infected. He was monitored in the hospital off antibiotics for >24 hours without recurrence fevers or nausea and vomiting. # Nausea with vomiting: He presented with acute onset of nausea and vomiting, although did admit to some chronic nausea since his esophagectomy. It was felt to most likely be caused by a viral gastroenteritis. Given his concurrent hypertension, he underwent head CT that showed ventriculomegaly and MRI was recommended. … ... … # Hypertension: His blood pressure was significantly elevated to 180/100's on admission and was very responsive to low doses of oral labetalol. He remained mildly hypertension (BP ___ during admission and was started on low dose lisinopril. He should have BP rechecked and electrolytes checked by his PCP and an appointment was arranged for him. # Transaminitis: AST and ALT mildly elevated on admission and remained mildly elevated. This may be related to his presumed viral syndrome. … ... … He should have LFTs rechecked at PCP ___. # Shoulder mass: He had a recent PET CT and MRI that showed likely metastatic disease in his shoulder. … ... … TRANSITIONAL ISSUES: - Hepatitis serologies, EBV, CMV pending - Needs repeat LFTs, - Started on lisinopril for HTN. Should have electrolytes checked and uptitration by PCP - MRI final read pending - Biopsy of shoulder mass scheduled for ___ - Blood cultures pending - CT abdomen incidentally showed bilateral basilar pulmonary nodules that are likely unchanged from prior scans. Recommend dedicated chest CT follow up in 6 months.

Fig. 2: A sample hospital discharge report (from MIMIC-IV dataset). We highlighted some primary reasons for the patient's admission, which may include fever and nausea, some important events during the hospital stay, which may include head CT or oral labetalol, and some essential follow-up actions, which may include chest CT and liver function tests (LFT).

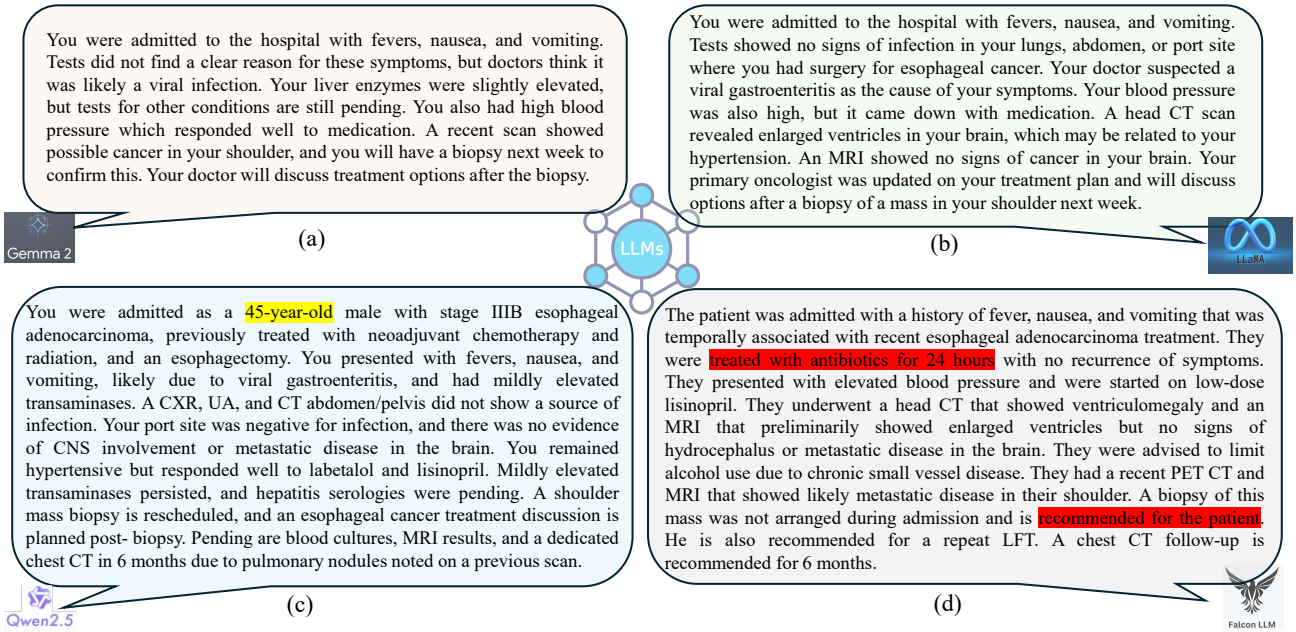

Fig. 3: Discharge report summary generated by (a) Gemma2, (b) LLaMA, (c) Qwen2.5 and (d) Falcon. Some instances of hallucinations are highlighted.

- (i) Unsupported facts [29]: The model generates clinical details, such as a diagnosis or prescription, that were not present in the original discharge summary. These can include unsupported conditions, procedures, medications, numbers, names and other details.
- (ii) Incorrect or Contradicted Facts [29]: The model misinterprets, distorts, or contradicts events, resulting in misleading conclusions that deviate from the original source and potentially alter the intended clinical meaning.
- (iii) Faithfulness hallucinations [30]: The model misses or omits key details of the source, altering the meaning of the summary and affecting its reliability.
- (iv) Content hallucinations [31]: The model introduces content that is completely unrelated or irrelevant to the source document.

In short, while some abstraction is expected in summarization, hallucinations introduce unverifiable or incorrect content. Using the discharge report in Fig. 2, we illustrate this with summaries from (i) Qwen2.5 and (ii) Falcon in Figs. 3(c)-3(d), respectively.

Firstly, the Qwen2.5 summary inaccurately assigns the patient an age of 45, a detail absent from the original report, exemplifying an unsupported fact hallucination. Additionally, it omits essential follow-up instructions for Liver Function Tests (LFTs), despite the report explicitly stating their necessity

twice. With 122 characters still available, this omission reflects a faithfulness hallucination, where the model fails to capture critical details vital for ongoing patient care.

In contrast, Falcon's summary inaccurately states that a biopsy "is recommended", whereas the original report confirms it was already rescheduled, misrepresenting the planned action. It also falsely claims the patient received antibiotics for 24 hours, while the report states they were monitored without antibiotics for over 24 hours, both errors falling under "incorrect facts hallucinations". Additionally, the summary inconsistently switches between 'they' and 'he' (e.g., "They were advised to limit alcohol..." vs. "He is also recommended for a repeat LFT".), causing gender hallucinations that compromise clarity.

IV. Experimental Results

In this section, we present the numerical experiments conducted using various open-source large language models (LLMs) applied to hospital discharge reports sourced from the MIMIC-IV dataset. Now we detail the experimental setup and discuss the outcomes derived from these evaluations.

A. Open-source LLMs and Dataset

In this study, we utilize open-source large language models (LLMs) to summarize hospital discharge reports. The selected LLMs include LLaMA, DeepSeek, and Phi etc. Table I provides an overview of these models along with their respective parameter counts. Here, we employ a subset of hospital discharge reports sourced from the publicly available MIMIC-IV dataset [29]. Each discharge report is sequentially processed using each LLM's standardized prompt.

> You are a helpful assistant that helps patients understand their medical records. You will be given some doctor's notes, and you will need to summarize the patient's brief hospital course in one paragraph (within 1000 characters). Please only include important and essential information and avoid using medical jargon, and you MUST start the summary with "You were admitted".

B. Key Information Extraction

We assess open-source LLMs on extracting (i) admission reasons, (ii) key events, and (iii) follow-ups from discharge reports, using GPT-4 as an evaluator [28]. Table I shows the percentage of key details captured, highlighting each model's strengths and limitations in medical text understanding.

From the results, Qwen2.5 and DeepSeek-v2 emerge as the top-performing models in extracting admission reasons, with Qwen2.5 achieving 83.33% comprehensively and 85% fairly. Similarly, Phi3 and DeepSeek-v2 show strong performance in capturing hospitalization events, while LLaMA3.1 and LLaVA exhibit relatively lower effectiveness. Follow-up recommendations prove to be the most challenging category for all models, with comprehensive coverage ranging from 29.33% (LLaMA3.1) to 55% (Phi3). However, fair coverage is comparatively better, with Phi3 and LLaVA achieving above 57%. These indicate that while LLMs perform reasonably well in extracting reasons for admission and key hospitalization events, their effectiveness in summarizing necessary follow-ups remains limited and requires further enhancement.

The observed limitations in extracting key events cannot be solely attributed to the upper bound of 1000 characters set for the summaries. As shown in Table I, many of the generated summaries fall well below this limit, yet they still fail to capture essential details such as reasons for admission, key hospitalization events, and necessary follow-ups. This suggests that the issue lies more in how the models prioritize information rather than in the imposed length constraint. Interestingly, some models, such as Phi3 and MistralLite, exceeded the given character limit, indicating that the restriction was not strictly enforced across all models. Their performance could improve with fine-tuning to better use available space and ensure comprehensive event extraction.

C. Hallucinations

Now, we examine unsupported facts and incorrect or contradicted fact hallucinations in LLM-generated hospital discharge summaries, which can impact their reliability. Unsupported fact hallucinations introduce details absent in the original report, while incorrect/contradicted fact hallucinations conflict with the source text. Table II summarizes the hallucinations identified across different LLMs from 100 discharge report summaries obtained from each of the LLMs.

The results reveal notable variations in hallucination tendencies among models. Phi3 exhibits the highest number of hallucinations in both categories, with 150 unsupported fact hallucinations and 111 incorrect/contradicted fact hallucinations, indicating significant reliability concerns. DeepSeek-v2, while generating fewer unsupported facts (46), shows a high number of contradicted facts (102), suggesting issues in accurately interpreting medical information. Other models, such as LLaMA3.1, Qwen2.5, and Gemma2, show more balanced yet still concerning levels of hallucinations, while MistralLite and LLaVA tend to generate a higher number of unsupported facts. These findings highlight the need for better fine-tuning and fact-verification to improve the factual reliability of LLM-generated medical summaries in clinical use.

V. Conclusion

Our study of open-source large language models (LLMs) for summarizing hospital discharge reports reveals their potential to accurately extract crucial clinical information but also highlights their susceptibility to generating hallucinations, including unsupported and incorrect facts. These inaccuracies pose risks to patient care and safety. Our comprehensive numerical simulations indicate variability in model performance, underscoring the need for model refinement to reduce hallucinations and enhance understanding of medical contexts. Future work can focus on fine-tuning models to enhance summary quality and mitigate hallucinations. Additionally, incorporating robust validation methods within clinical workflows can improve the reliability of LLM-generated summaries, providing healthcare professionals with efficient and accurate tools for patient care.

TABLE I: Comparison of LLMs in terms of the total number of extracted key events from the respective summaries. While LLMs perform quite well (up to 83.33%) to find out the reasons, their performance is limited to figuring out the follow-ups (can be as low as 29.33%).

LLM	# Parameters	Avg. Length (SD)	"Comprehensively" Covered			"Fairly" Covered		
			Reasons	Events	Follow-up	Reasons	Events	Follow-up
LLaMA3.1	$8B$	674(95)	70.33%	58%	29.33%	71.67%	59.67%	30.33%
Qwen2.5	$7B$	778(160)	83.33%	70%	50.67%	85%	70.67%	52.33%
DeepSeek-v2	$16B$	1010(332)	80.33%	65%	41.33%	81%	67.33%	42.67%
Phi3	$3.8B$	1695(580)	81.33%	63%	55%	83%	65.67%	58.33%
Gemma2	$9B$	621(124)	69.67%	55.67%	30%	70.67%	58.33%	31.33%
MistralLite	$7B$	1400(795)	77.67%	69%	49%	78.67%	70%	50%
LLaVA	$7B$	993(416)	70.67%	61.67%	53.33%	73.33%	64.33%	57.33%

TABLE II: Comparison of LLMs in terms of total number of hallucinations within the considered 100 summaries.

LLM	Total number of Unsupported fact Hallucinations	Total number of Incorrect/Contradicted fact Hallucinations
LLaMA3.1	85	27
Qwen2.5	92	30
DeepSeek-v2	46	102
Phi3	150	111
Gemma2	75	25
MistralLite	86	43
LLaVA	85	37

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